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Analysis of feature extraction from protein sequences

Building a pipeline for protein feature extraction

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Foreword

As I present this thesis, I am deeply grateful for the support and guidance that have shaped my journey in the complex field of protein structure analysis. I extend my sincere appreciation to dr. ir. Pieter Coussement, my mentor, and thesis supervisor, for his invaluable guidance in navigating the intricacies of biology and reviewing my work. My thanks also go to Medha Hegde and Sharon Grundmann from ML6, whose expertise and feedback on the data science aspects of my internship were invaluable.

I am grateful to Frédéric Vlummens, my internship coach at Howest, for his unwavering support throughout this journey. I extend my heartfelt thanks to ML6 for providing me with the opportunity to undertake my internship, which greatly enriched my understanding and skills in protein structure analysis.

Lastly, I am indebted to my family for their unwavering emotional support.

To everyone who supported me along this path, thank you for your guidance, encouragement, and belief in my abilities. Your contributions have been invaluable to this thesis.

Warm regards,

Denis Topallaj, Bruges, 03/06/2024

Abstract

All medicine is produced using fossil fuels. In the face of climate change, a need for sustainable production methods has never been clearer. With ecosystems under strain and traditional industries contributing to environmental degradation, there is a pressing need to rethink how we produce goods and utilize resources.

The quest for sustainable production methods has led to the development of industrial biotechnology, leveraging microbial cell factories (MCFs) to convert renewable feedstocks, such as sugar, into high-value products. Enzymes play a crucial role in optimizing MCF performance, but their discovery and design can be a time-consuming and labour-intensive process.

This thesis proposes a pipeline for the automated calculation of enzyme features, enabling the efficient design and optimization of enzymes for all kinds of biotechnology applications. By integrating cutting-edge tools for protein feature calculations, this pipeline aims to accelerate the discovery of high-performance enzymes for the sustainable production of valuable compounds like flavonoids and terpenoids.

Keywords: climate change, enzymes, microbial cell factories, feature extraction, flavonoids, terpenoids

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Glossary

- **acidity or alkalinity:** pH measure
- **amino acids:** building blocks of proteins
- **AI models:** artificial intelligence systems for various tasks
- **alpha-helices:** common protein secondary structure
- **beta barrels:** protein structural motif
- **beta sheets:** protein secondary structure
- **beta turns:** protein structural element
- **caching:** storing data in a temporary storage area
- **canalization:** genetic robustness to environmental variation
- **check sum:** a value used to verify data integrity
- **cytochrome P450 enzyme:** protein catalyst for accelerating chemical reactions
- **dataframe:** a data structure in programming for organizing data
- **deCYPher:** biology and AI project founded by the European Commission
- **descriptors:** attributes used to describe data
- **Docker:** a platform for developing, shipping, and running applications
- **Docker containers:** run isolated software applications, providing a lightweight and portable environment for development, testing, and deployment across various platforms
- **DNA (deoxyribonucleic acid):** genetic material. Double stranded biological blueprint for life, existing of deoxyribonucleotides
- **electrical charge:** property of particles
- **ensemble method:** using multiple machine learning models to improve performance
- **feature:** an individual measurable property or characteristic
- **feature engineering:** creating input features for ML models
- **Fondant:** a data processing framework created by ML6
- **GitHub:** a platform for hosting and collaborating on code
- **Hugging Face:** a French American AI, datasets … hosting and serving website
- **large language model fine-tuning:** adapting pre-trained models to specific tasks
- **learning rate:** it controls the step size or rate at which the model learns from each iteration
- **least squares regression:** A statistical method for estimating the coefficients of a mathematical model that minimizes the sum of the squared differences between observed and predicted values
- **lipid bilayer:** structure forming cell membranes
- **machine learning:** a field of AI where systems learn from data
- **MIT license:** a permissive software license allowing broad use with minimal restrictions
- **ML6:** an AI consultancy company, where this internship has been done
- **MSA:** multiple sequence alignment in bioinformatics
- **overfitting:** occurs in machine learning when a model learns to capture the noise in the training data rather than the underlying pattern or relationship between variables
- **oxygenation:** process of adding oxygen to a molecule
- **pandas:** a data manipulation library in Python
- **parameter tuning:** the process of adjusting the parameters of a machine learning algorithm to optimize its performance on a given dataset
- **PDB:** protein data bank. A format in which tertiary protein sequences are written as
- **pH:** measure of acidity or alkalinity (potential of hydrogen)
- phenylalanine, tyrosine, tryptophan: amino acids
- **pipeline:** a sequence of data processing components
- **POC:** proof of concept. A small-scale project to show if an idea works before investing fully.
- **primary sequence:** the linear sequence of amino acids of a protein

- **prokaryotes:** a single-celled organism lacking a cell nucleus and organelles, like bacteria
- **protein:** biological molecules essential for life
- **PyPI**: python package index. A wide range of packages used in Python
- **Python:** a popular programming language
- **regularisation:** techniques used to prevent overfitting by adding a penalty term to the model's objective function
- **RNA (ribonucleic acid):** short-lived, single-stranded genetic material existing out of deoxyribonucleotides. Transcribed from DNA.
- **RMSE (Root Mean Squared Error)**: A measure of the average deviation of predicted values from actual values, taking the square root of the average of the squared differences.
- **secondary sequence:** local folding patterns in proteins
- **solubility:** ability of a substance to dissolve
- **tertiary sequence**: overall 3D structure of a protein
- **TMD:** transmembrane domain in proteins
- **tyrosine**: amino acid
- **underfitting:** occurs when a machine learning model is too simple to capture the underlying structure of the data
- **UniProt:** a comprehensive website for providing information on proteins, including their sequences, functions, and structures

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1 Introduction

To mitigate climate change, adopting sustainable production methods is crucial. One approach is to utilize renewable resources as feedstock, such as sugar, for industrial processes. Industrial biotechnology converts these feedstocks into high-value products using microbial cell factories (MCFs). Effective enzymes are essential for optimizing MCF performance, which can be naturally occurring or engineered. The integration of industrial biotechnology with renewable feedstocks offers a promising solution for sustainable production, reducing environmental impact and providing a competitive edge through high-value products.

Enzymes are proteins that act as biological catalysts by accelerating chemical reactions in living organisms. They are essential for various biological processes, including digestion, metabolism, and the breakdown of nutrients. Enzymes are highly specific, interacting with only one type of substrate or group of substrates to catalyse a specific reaction.

To find or design these enzymes, Design Build Test Learn (DBTL) cycles are often used. By maximizing the learnings from previous experiments, suggesting new ones. One essential step in the DBTL cycle for the development of enzymes is the automated calculation of features.

Currently, tools like Biopython, iFeatureOmega and other which calculate the features of primary protein sequences do exist. However, 3D structures are also very important. Tools such as ESMFold, AlphaFold and other can predict the tertiary structure of protein sequences. [1] [2] [3]

This thesis focuses on the building of a pipeline that can calculate features in a flexible and collaborative way. It also fits into the deCYPher project, which is a specific case study for the sustainable production of flavonoids and terpenoids, which are valuable compounds with various applications in the pharmaceutical and food industries. [4]

1.1 The complexity of protein sequences

Enzymes work by lowering the activation energy needed for reactions to occur, making processes more efficient. Each enzyme has a specific function, breaking down or building up substrates into products. Enzymes have an ideal temperature and pH range for optimal function, and their activity can be influenced by factors like inhibitors or activators. [5]

Protein sequences can be classified into different types based on their structure and organization. The four main levels of protein structure are primary, secondary, tertiary, and quaternary.

Primary structure: the primary structure of a protein is the exact ordering of amino acids forming their chains. A change in the amino acid sequence can affect the protein's overall structure and function, as seen in genetic disorders such as cystic fibrosis [6], which is caused by mutations resulting in alterations in the primary protein structures (of DNA).

Secondary structure: the secondary structure of a protein refers to the local folded structures that form within a polypeptide due to interactions between atoms of the backbone. These local folded structures are found in two different types: α-helix and β-pleated sheet structures. The α-helix structure is formed by hydrogen bonds between carbonyl and amino groups that make up the polypeptide backbone, causing the molecule to spiral around. The β-pleated sheet structure is formed by hydrogen bonds between carbonyl and amino groups that make up the polypeptide backbone, causing the molecule to bend and fold.

Tertiary structure: the tertiary structure of a protein is formed by many different bonds between R groups that make up the side chains, causing the strand of molecule to bend and loop around in a more complicated three-dimensional form. The tertiary structure is crucial for the protein's functionality, as it determines the specific positioning of different regions of the protein, affecting its ability to interact with other molecules, substrates, and parts of the cell.

Quaternary structure: the quaternary structure of a protein is formed when multiple folded polypeptide chains associate to form a complex functional protein. The association of folded polypeptide molecules to complex functional proteins results in quaternary structure, which is important for the protein's functionality.

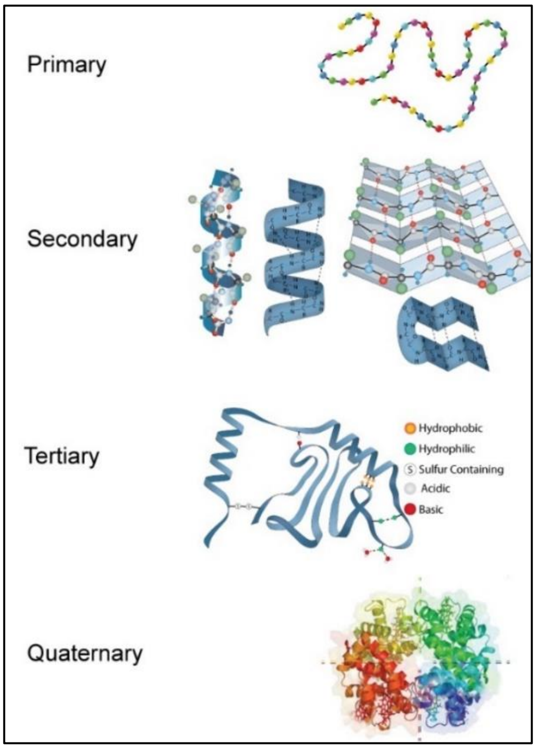

Figure 1: Types of protein structures [7]

1.2 Feature Calculation

Feature extraction from protein sequences is essential for understanding protein characteristics and plays a vital role in drug discovery. Enzymes like CYP enzymes are crucial for drug development and extracting features from them helps uncover catalytic mechanisms and design new drugs. This process also reveals common patterns and structural elements in proteins, providing insights into their functions and history. Customized feature extraction methods aid in protein engineering for tailored traits. Projects like deCYPher aim to place enzymes like CYP into fast-growing bacteria for various applications in medicine, cosmetics, and industry. This innovative approach leverages bacteria's rapid reproduction to address diverse challenges in different sectors.

There are different ways to calculate features from protein sequences.

- **Global Features:** These include sequence length, molecular weight, isoelectric point, and other overall properties that describe the entire protein sequence.
- **Per Amino Acid Features:** This category involves features specific to each amino acid, such as amino acid composition, hydrophobicity index for individual amino acids, and other properties that focus on the characteristics of each amino acid in the sequence.
- **Between Amino Acids Features:** Features in this group capture relationships and interactions between amino acids in the sequence. This can include similarity and distance metrics between sequences, highlighting patterns and connections between different amino acids within the protein sequence.

These diverse types of features provide a detailed and multi-dimensional view of protein sequences, enabling researchers to analyze and understand the complex nature of proteins effectively.

1.3 Research question

Extracting meaningful information from protein sequences is crucial for understanding how proteins work. Feature extraction techniques help us do this, enabling us to develop algorithms that can predict protein structures and functions. To get the most accurate results, we need to make sure our feature extraction methods are optimized and can handle different types of protein sequences. This helps us better understand the different structures and functions of proteins.

With this background, the research questions are the following: How can we extract the most useful features from protein sequences?

The question above can be divided into sub-questions:

- How can feature extraction techniques be optimized to capture relevant information from protein sequences in an automated and collaborative way?
- What machine learning algorithms are most effective for predicting protein structures based on extracted features?

- How can the predictive accuracy of the feature extraction pipeline be validated and improved?
- How can feature extraction techniques be adapted to accommodate variations in protein sequence data, such as length and complexity?

Throughout this bachelor thesis, the various methodologies and techniques aimed at extracting the most informative features from protein sequences will be explored. By addressing this research question, the thesis aims to contribute to the advancement of computational tools and methods for protein analysis, ultimately providing insights that can aid in diverse applications.

2 Literary study

This section will contain the approach and design of the project together with discussions of papers.

Given that this thesis is for a bachelor's degree in Applied Computer Science rather than biology, guidance from dr. ir. Pieter Coussement (internship mentor) was sought after for clarifications on biology-related concepts. These consultations covered fundamental questions such as the definition of a protein sequence and methods for modifying primary protein sequences to different types. This collaborative approach ensured a comprehensive understanding of the biological underpinnings relevant to the research, facilitating informed decision-making throughout the study.

2.1 Data collection

The protein data comprises primary sequence proteins organized in a tabular format, where each row represents a specific protein, and each column corresponds to different features. The dataset collected for the proof of concept is obtained from the UniProt website. [8]

2.2 Importance of feature extraction

Feature extraction is a crucial step in protein sequence analysis, where relevant characteristics of the protein are identified and represented in a structured format. The features are needed to create a regression model trained on these features. This model will be able to accurately predict a feature using new protein sequences. Many tools are described in literature and there are public tools available for feature extraction.

After extracting features from the protein sequences, it becomes imperative to assess their efficacy in encapsulating the structural and functional attributes of the proteins. Evaluating these features commonly entails comparing them against established experimental data or functional annotations.

In this project, the dataset will be partitioned into training and testing subsets. The testing subset will be utilized to predict the correct prediction feature[1] based on the derived features. To evaluate the performance of the features, a regression model will be employed. This model will provide insights into how well the extracted features align with the desired predictive outcome, thereby validating their utility in capturing the desired structural and functional characteristics of the proteins. What the feature that will be predicted is, will be discussed further in the thesis.

2.3 Tertiary structure prediction

There are various possible models utilized for predicting the intricate three-dimensional arrangement of proteins. The most recent models for these predictions are ESMFold and AlphaFold2. The two models will be discussed in this section.

2.3.1 ESMFold

ESMFold is an artificial intelligence method for predicting tertiary protein structures. It utilizes the ESM-2 language model to generate accurate structure predictions directly from the sequence of a protein. This ESM-2 model is pretrained using a Masked Language Model for protein sequence. This means that certain amino acids are hidden from the protein sequence and the task of the model is finding out what these missing parts are.

ESMFold is designed to predict atomic-level protein structures with high accuracy, like other advanced tools like AlphaFold2. This method enables end-to-end structure prediction, offering fast inference times compared to traditional methods. [9]

The main difference between ESM-2 and ESMFold is that ESM-2 is a language model for protein sequences that captures a more accurate picture of protein structure even at 150M parameters, while ESMFold is a protein structure prediction model that uses the ESM-2 language model to enable high-accuracy end-to-end atomic-level structure prediction directly from individual protein sequences, offering faster prediction speeds and the ability to explore structural space efficiently. [10]

[1] The feature that will be used in the regression model to predict outcomes

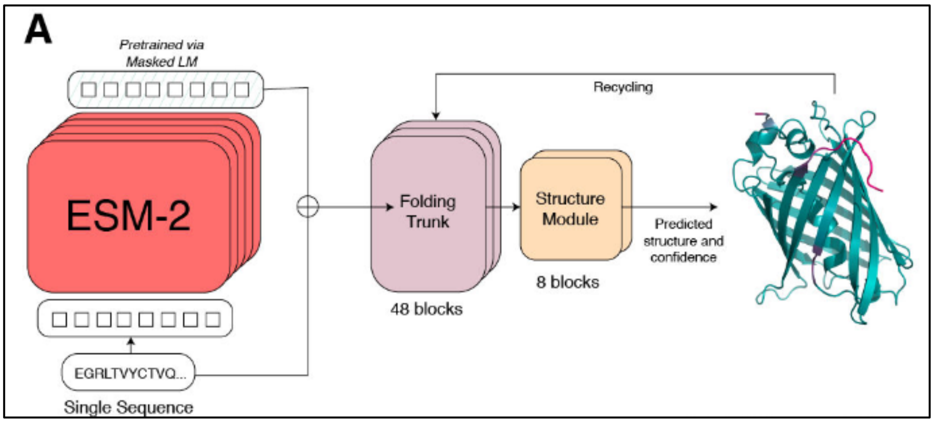

Figure 2: ESMFold model architecture [9]

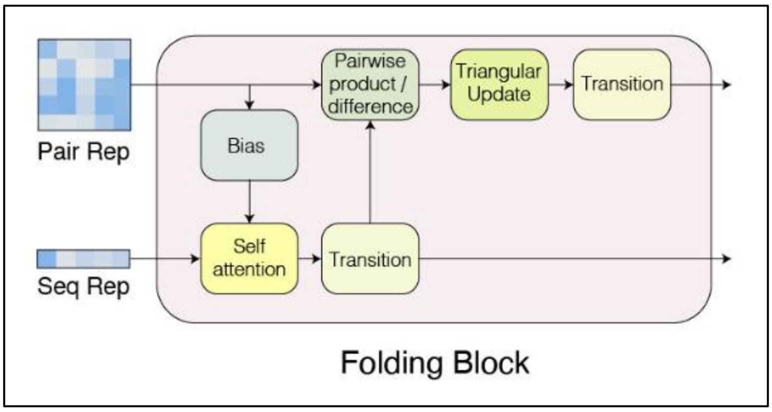

Figure 3: Folding Block of the ESMFold architecture [9]

The figures above showcase the architecture of the ESMFold model. The folding block takes the output from the language model and processes it to generate a sequence representation that captures the evolutionary information about the protein. The folding trunk consists of multiple folding blocks, each of which alternates between updating a sequence representation and a pairwise representation.

The output from the folding trunk is then passed to the structure model, which is designed to generate a 3D atomic-level structure from the sequence representation. These two steps are recycled. The number of recycling steps determines how many times the model refines its prediction by building upon its previous output. [9]

2.3.2 AlphaFold

AlphaFold, developed by DeepMind [11], is a protein structure prediction model that uses a neural network-based approach to predict the three-dimensional structure of a protein from its amino acid sequence. This model is designed to predict atomic-level protein structures with high accuracy for a wide range of proteins. AlphaFold uses a combination of bioinformatics and physical approaches to build components that learn from protein data with minimal imposition of handcrafted features. This results in a network that learns efficiently from limited data in the Protein Data Bank (PDB) but can handle the complexity and variety of structural data.

AlphaFold is trained to produce the protein structure most likely to appear as part of a PDB structure, incorporating novel neural network architectures and training procedures that significantly improve the accuracy of structure prediction. The model outputs protein coordinates directly, allowing for fast prediction speeds and the ability to explore structural space efficiently.

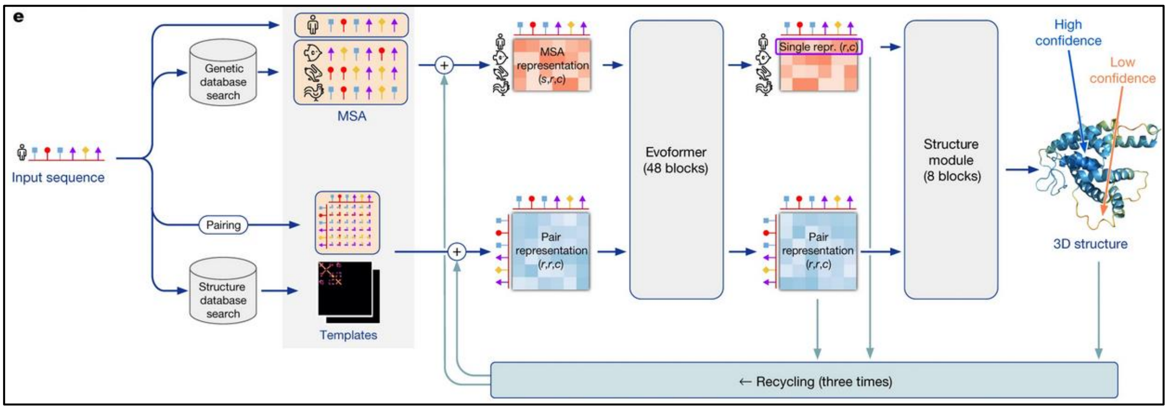

Figure 4: AlphaFold model architecture [12]

The image above illustrates the model's architecture. Firstly, the input module takes in amino acid sequence as input and generates a multiple sequence alignment (MSA) representation and a pair representation. The MSA identifies similar sequences and allows detection of correlated mutations. The pair representation models which amino acids are likely to be in contact.

The Evoformer module is up next. This module consists of 48 blocks that iteratively update the MSA and pair representations using a deep learning architecture. Each block contains components that model evolutionary couplings, attention between sequence features, and structural motifs.

The structure model converts the abstract representations from the Evoformer into 3D atomic coordinates of the protein structure. It takes as input the single representation (a linear projection of the MSA), the pair representation, and backbone frames representing the positions of the amino acid backbone atoms. The structure module contains 8 blocks that refine the structure prediction.

2.3.3 Comparing ESMFold and AlphaFold for Protein Structure Prediction

When selecting between ESMFold and AlphaFold for protein structure prediction several key factors should be considered:

- **Accuracy:** AlphaFold2 is currently the most accurate method for protein structure prediction, outperforming ESMFold and other models [13]
- **Speed:** ESMFold is reported to be 60 times faster than AlphaFold2 for short proteins, although this speed difference is less pronounced for longer sequences. This is due

to to the fact that ESMFold doesn't use MSA, which slows down the prediction and folding. [3]
- **Input requirements**: AlphaFold2 requires multiple sequence alignments (MSAs) as input to leverage evolutionary information, while ESMFold uses only the single protein sequence. [12] [3]
- **Memory usage**: ESMFold has higher memory requirements compared to AlphaFold2. [13]

The choice within the project has been made to use ESMFold for its fast prediction and minimal input requirements.

3 Experiment

This section will contain a comprehensive overview of the experiment. This experiment has been performed during the internship with the help of ML6 and deCYPher.

3.1 Methodology

The methodology of this project consists of using digital, publicly available tools for biotechnology. Creating a pipeline using Fondant [14] that can be used by anyone with collaborative pipeline building. The dataset of features, extracted from the sequences, will be used in a regression model to predict the next suitable protein for testing.

The deCYPher project uses a Design-Build-Test-Learn (DBLT) cycle where data created during the cycle, will be used in the next cycle as input. This project situates itself between the Test and Learn section of the DBLT cycle. [4]

3.1.1 Fondant

Fondant, a collaborative data processing framework developed by ML6, is designed to enhance, and streamline extensive data processing operations. It emphasizes flexibility and modularity, offering a range of features like easily combinable data processing pipelines, a collection of reusable components, a Pandas-based interface for personalized components, integrated lineage tracking, caching capabilities, a data explorer, and more. [14]

Fondant provides a standardized way to build and combine components for creating datasets for AI models, including image generation, large language model fine-tuning, and code generation model fine-tuning. It emphasizes the reusability of containerized components across pipelines and execution environments, making them shareable within the community.

Components can work independently of each other. This framework also includes publicly available components that perform certain repetitive work, such as loading the dataset, cropping images, loading data from other sources and much more.

Fondant uses Docker under the hood, where each component is turned into a Docker Container. These Docker Containers can be easily shared to other people with similar datasets.

Fondant also has a wide range of possibilities for integration of cloud code runners. The pipeline can be easily transferred to different cloud providers, by choosing between Google's Vertex AI, AWS Sagemaker and Google's Kubeflow. These cloud providers can be interchanged easily and do not require changing of big parts of code, which removes vendor lock-in[2] as an option. [14]

[2] Vendor lock-in refers to a situation where a customer becomes dependent on a particular vendor's products or services, making it difficult to switch to alternatives.

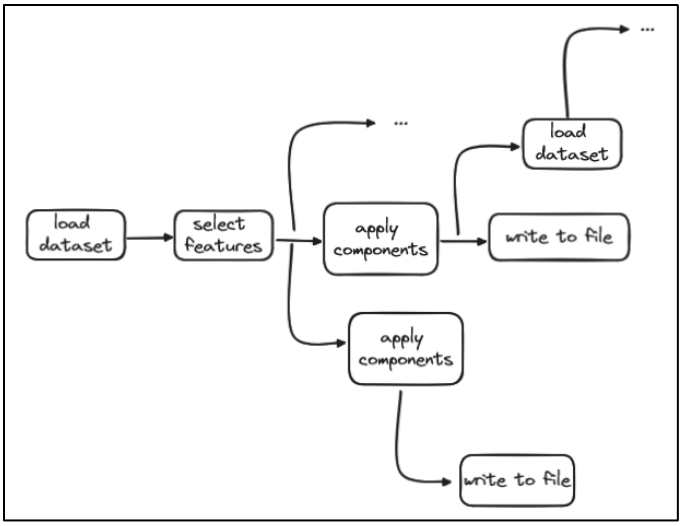

Figure 5: Example Fondant Structure

The image above shows a possible Fondant pipeline that could be implemented. The Fondant framework does not have to be linear and can be used for a wide range of operations at the same time.

3.2 Components

This section will contain all the components that are used in the pipeline to extract features in different ways. Each component contributed uniquely to the feature extraction process, enhancing the overall methodology and outcomes. By dissecting and analysing these components individually, a deeper understanding is known of how they synergistically come together.

3.2.1 Biopython Component

Biopython, a collection of freely available Python modules for computational molecular biology, offers a versatile set of tools for bioinformatics applications. It also allows users to access parsers for various bioinformatics file formats, online services, sequence manipulation tools, and phylogenetics analysis modules.

Biopython, a specialized bioinformatics tool, offers a diverse range of features tailored for biological data analysis. It includes parsers for genetic databases, tools for protein structure management, and access to online services like NCBI and ExPASY. Developed by a global community since 2000, Biopython stays current with bioinformatics advancements. Known for its user-friendly syntax and extensive documentation, Biopython seamlessly integrates with bioinformatics resources, making it accessible and efficient for researchers of all levels. [15]

The features that have been chosen and calculated using this component:

- **Sequence length:** This feature simply calculates the length of the protein sequence. It provides basic information about the size of the protein.
- **Molecular weight:** Molecular weight calculates the total mass of the protein sequence by summing the masses of all its constituent amino acids. It is crucial for understanding the physical properties and behaviours of proteins.
- **Aromaticity:** Aromaticity measures the fraction of amino acids in the sequence that are aromatic (such as phenylalanine, tyrosine, and tryptophan). Aromatic amino acids play important roles in protein structure, stability, and function.
- **Isoelectric point:** The isoelectric point is the pH at which a protein carries no net electrical charge. It is a critical parameter that influences protein solubility, stability, and interactions with other molecules.
- **Instability index:** The instability index predicts the stability of a protein based on its amino acid composition. Higher values indicate higher instability, suggesting potential regions prone to unfolding or degradation.
- **Gravy:** GRAVY (Grand Average of Hydropathicity) calculates the hydropathy of a protein sequence, representing the sum of hydropathy values of all amino acids divided by the number of residues. It provides insights into the overall hydrophobicity or hydrophilicity of the protein. Hydropathy is a boarder term that consists of hydrophobicity (water repelling) and hydrophilicity (water attracting) properties.
- **Helix, turn, sheet:** These features predict the propensity of the protein sequence to form α-helices, β-turns, and β-sheets, respectively. They are crucial for understanding protein secondary structure and folding.
- **Charge at pH 3, 5, 7 and 9:** These features calculate the net charge of the protein at their respected values. Understanding the protein's charge distribution is important for predicting its behaviour in different cellular environments and interactions with other molecules. pH affects protein sequences and amino acids by altering the charge of individual residues, which can lead to changes in protein structure, function, and interactions. This is because the charge of amino acids is pH-dependent, with some amino acids becoming charged or uncharged at different pH levels. [16]
- **Molar extinction coefficient oxidized, and molar extinction coefficient reduced:** These coefficients provide information about the protein's absorbance properties at specific wavelengths, particularly when the protein is in its oxidised or reduced state. They are important for protein quantification and characterization in biochemical assays.
- **Flexibility:** Calculates the maximum, minimum, and mean flexibility values for a protein sequence, reflecting the structural flexibility and potential for conformational changes.

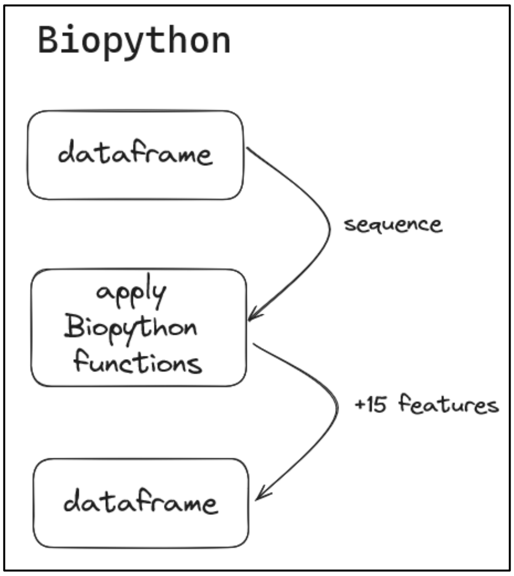

Figure 6: Biopython Component Structure

3.2.2 iFeatureOmega Component

The iFeatureOmega platform, developed by the Kurgan Lab, is a versatile tool for generating, analysing, and visualizing a wide array of representations for biological sequences, 3D structures, and ligands. iFeatureOmega is a recent package that was created around Dec 8, 2021. [17]

It provides over 180 descriptors for biological sequences, structures, and ligands, making it a valuable resource for bioinformatics research. Descriptors are names that are used within the project of iFeatureOmega to group certain features together.

The platform is implemented using Python, with support for CLI, GUI, and web server versions. It leverages third-party software packages like Biopython, Pandas, Numpy, RDKit, and scikit-learn for various functionalities. iFeatureOmega can generate 73 features using protein sequences, but most of these the final pipeline does not use. [2] A selection has been made to determine what features are important. This list has been chosen by Pieter Coussement.

The features that have been chosen and calculated using this component:

CTD stands for "Composition/Transition/Distribution" and is used as the boarder term to combine the Composition, Transition and Distribution descriptors together.

- **AAC (Amino Acid Composition):** The Amino Acid Composition (AAC) encoding calculates the frequency of each amino acid type in a protein or peptide sequence.
- **CTDC (Composition descriptor of CTD):** CTDC focuses on hydrophobicity, categorizing amino acids into polar, neutral, and hydrophobic groups to calculate the global composition of these residues in a protein.
- **CTDT (Transition descriptor of CTD):** CTDT, on the other hand, examines transitions between these groups, calculating the frequency of transitions like polar to neu-

tral or neutral to polar residues. These descriptors have been successfully applied in various bioinformatics tasks such as protein folding class prediction, enzyme family classification, and protein structural prediction. [18]

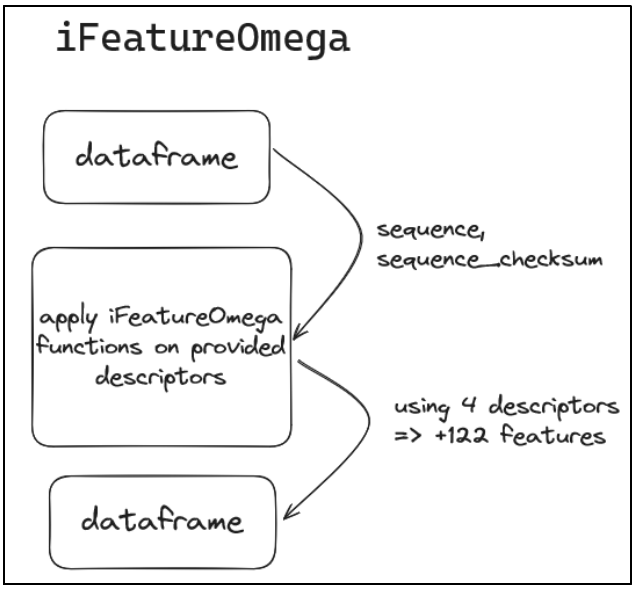

Figure 7: iFeatureOmega Component Structure

3.2.3 PDB Components

The Protein Data Bank (PDB) is a global repository that archives the three-dimensional structural data of large biological molecules, such as proteins and nucleic acids. The PDB format is widely used in computational biology for storing and analyzing the three-dimensional structures.

The PDB Component plays a vital role in managing and generating protein structure files in the PDB format. It facilitates the interaction with cloud or local resources to streamline the process efficiently.

The three-dimensional shape of the protein directly influences its function and interactions with other molecules. Identifying active sites and comprehending how proteins interact with ligands or other proteins is provided by the tertiary structure. Additionally, the tertiary structure provides insights into the stability, folding, and dynamics of proteins, which are vital for studying protein-protein interactions, enzymatic activities, and molecular mechanisms in biological systems.

Data Management and Retrieval

- The component establishes connections either with the Google Cloud Platform (GCP) infrastructure or local storage for storing PDB files, ensuring seamless access to resources for efficient data management and retrieval.

Checksum Operation and PDB Structure Generation

- It performs checksum operations on protein sequences to avoid redundant PDB file generation. If a match is found, indicating an existing PDB file, redundant computation is eliminated. Otherwise, it uses state-of-the-art tools to predict and generate new PDB structures.

Existing and Newly Generated PDB Integration

- For existing PDB files, their content is retrieved and merged with the original dataframe. Newly generated PDB files are integrated into the dataset, combining existing and predicted structures for comprehensive analysis.

Dataframe Management

- The component updates the dataframe with both existing and newly generated PDB structures, serving as the foundation for downstream analysis and processing within the pipeline.

The PDB component both predict the PDB files with or without a cloud provider. The PDB file will be represented as a string inside a column. This column can be useful for other tasks, but it will not be used in the regression model, since regression models only take in numbers and not letters.

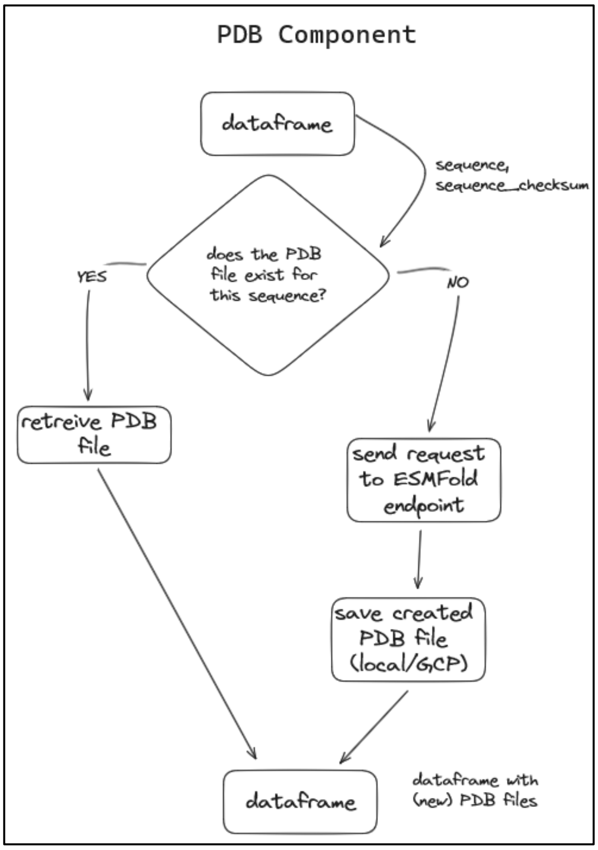

Figure 8: PDB Component Structure

To predict these tertiary sequences in the pipeline the ESMFold model has been placed on Hugging Face. Using the Interface Endpoints from Hugging Face, the endpoint can use the ESMFold model. This endpoint receives primary structure protein sequences and returns a tertiary structure protein sequence in a PDB format. This PDB file is then placed in the dataframe as mentioned above.

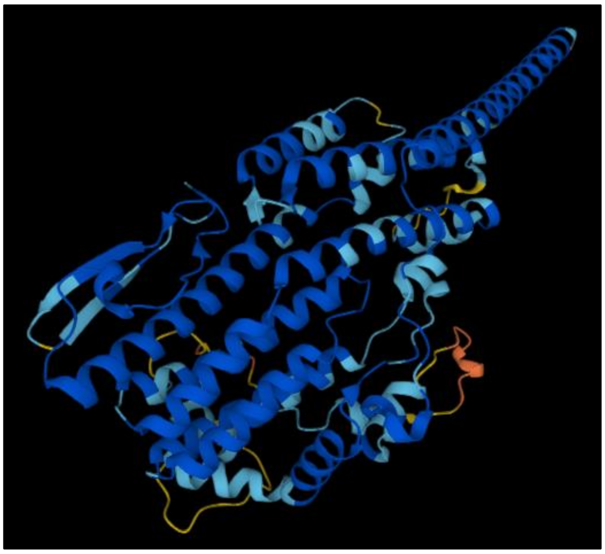

Figure 9: Tertiary structure of a protein sequence predicted by ESMFold [3]

3.2.4 MSA Component

Multiple sequence alignment (MSA) is a fundamental bioinformatics technique that aligns three or more biological sequences, typically protein, DNA, or RNA sequences. The primary objectives of MSA are to identify regions of similarity or conservation across the sequences, which can highlight homologous features and reveal evolutionary relationships, as well as to pinpoint mutation events such as point mutations (single amino acid or nucleotide changes), insertion mutations, and deletion mutations. By assessing sequence conservation, MSA allows researchers to infer the presence and activity of protein domains, tertiary structures, secondary structures, and individual amino acids or nucleotides.

MSA is necessary for calculating features between amino acids because it aligns the sequences and identifies conserved regions, ensuring that amino acids at the same position are compared across multiple sequences. This alignment process allows for the calculation of various features, such as amino acid frequencies at each position, coevolution scores between pairs of positions, and sequence conservation scores at each position.

Publicly available models are used to predict the MSA of protein sequences. The most notable are: Clustal Omega, MUSCLE, T-Coffee, HMMER, Kalign and much more.

Clustal Omega has been chosen for the use of MSA on this project, given that it’s the most accurate program, but it’s certainly not the fastest. A tradeoff has been made to prefer accuracy of time complexity. [19]

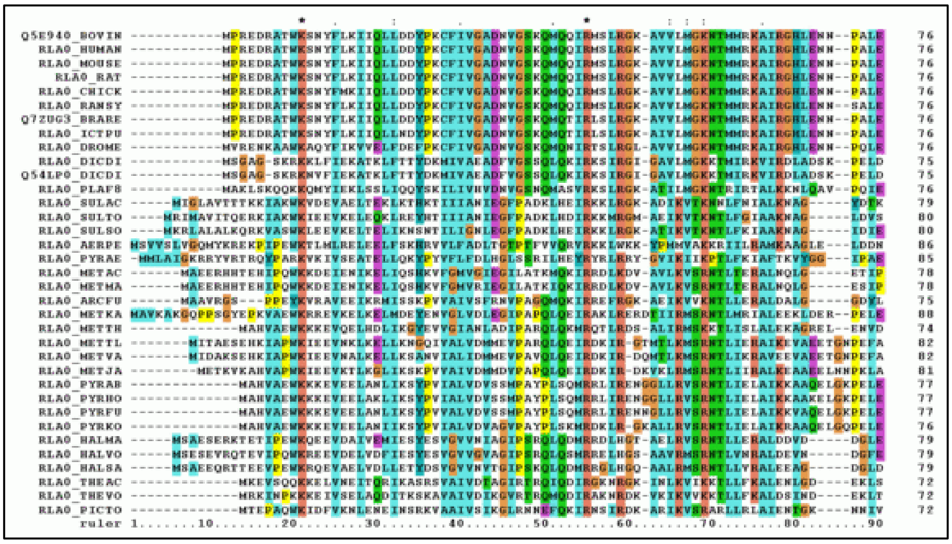

Figure 10: Protein sequences aligned by MSA [20]

The figure above is an example of how protein sequences would be aligned using MSA. The color shows the correlation between the same amino acids. To make the sequences aligned, dashes are used as padding to move the right amino acids in the corresponding place.

However, it is important to note that because MSA requires a large amount of protein sequence data to create a result, it limits the possibility for parallelization, as all the sequences need to be processed together to identify conserved regions and align the sequences accurately.

In other words, the more protein sequences utilized for feature extraction, the longer the process takes due to the limitations of parallelization in MSA. However, this extended processing time results in more accurate alignments as MSA aligns with a larger dataset, revealing additional patterns and enhancing the visibility of conserved regions. Conversely, using less data may speed up the process, but could compromise the accuracy and depth of the alignment and pattern recognition achieved through MSA. [20]

The MSA component uses the Clustal Omega program to predict the MSA structures using the sequences. To perform multiple sequence alignment on the protein sequences, all the sequences fed into the program at once. The result is a single file with all the sequences combines containing padding to match the length of the largest sequence. These aligned sequences will be placed inside the dataframe.

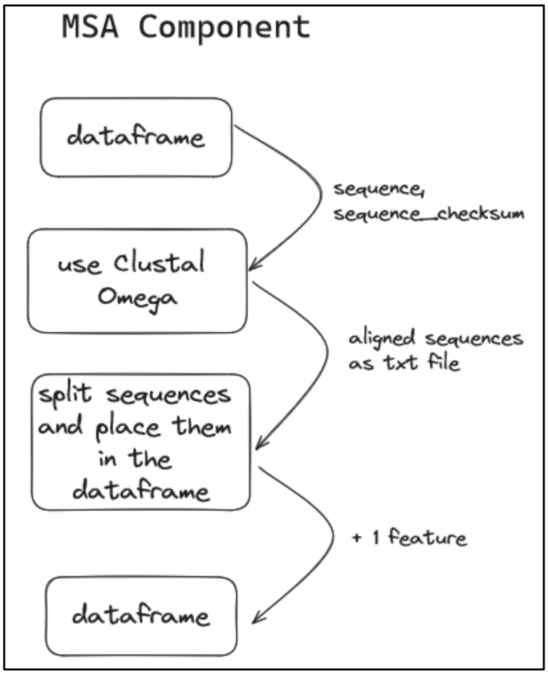

Figure 11: MSA Component Structure

3.2.5 TMD Component

A transmembrane domain (TMD) is a protein domain that spans the lipid bilayer of a membrane.

Membranes are essential components of cells, serving as barriers that separate the interior of the cell from the external environment and compartmentalize the cell into specialized organelles.

Membranes exist in various forms throughout the cell:

1. The plasma membrane surrounds the entire cell, enclosing its contents and controlling the movement of substances in and out of the cell.
2. Organelles, such as the nucleus, mitochondria, and endoplasmic reticulum, are also enclosed by membranes.
3. Membranes are composed of a lipid bilayer, with hydrophilic heads facing the aqueous environments on both sides and hydrophobic tails forming the interior of the membrane.

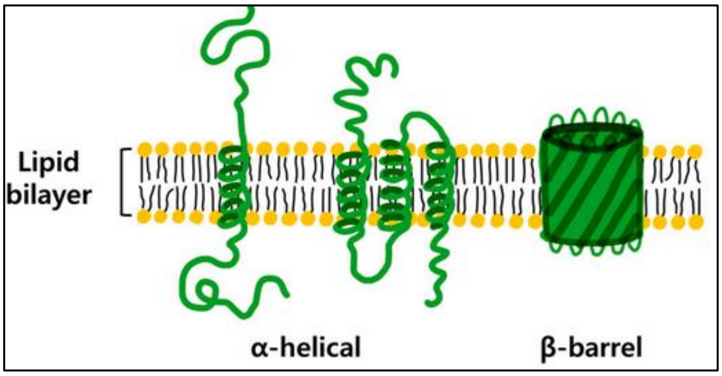

Figure 12: Alpha helices and Beta barrels [21]

TMDs exist because proteins need to interact with and traverse these membranes to perform various functions. The amino acid residues in TMDs are often hydrophobic to match the hydrophobic interior of the lipid bilayer. However, some proteins, such as membrane pumps and ion channels, may contain polar residues within their TMDs to facilitate the transport of specific molecules or ions across the membrane. [21]

The TMD Component uses DeepTMpred, which is a framework that predicts the alpha-helical transmembrane proteins. [22]

Transmembrane domains are crucial for feature extraction of protein sequences because they play a vital role in protein structure and function. Identifying these domains helps in understanding how proteins interact with cell membranes, which is essential for various cellular activities like signal transduction and transport. Analysing transmembrane domains aids in predicting protein topology, determining protein localization, and studying protein-protein interactions. This information is valuable for bioinformatics research as it provides insights into the structural characteristics of proteins that are integral to their biological functions. [23] [24]

The TMD component generates features such as the amount, total length, average, longest, and shortest length of helices.

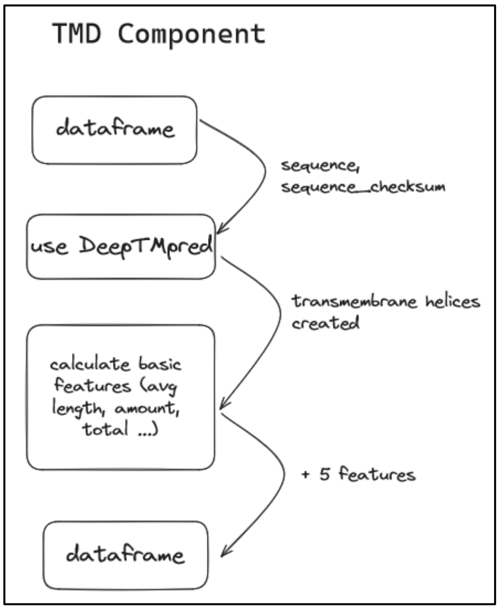

Figure 13: TMD Component Structure

3.2.6 UniKP Component

UniKP stands for Unified Kinetic Parameter prediction, a cutting-edge framework that leverages pretrained language models to predict essential enzyme kinetic parameters from protein sequences and substrate structures. This comprehensive framework encompasses multiple models and self-pretrained models, amalgamating their capabilities to predict crucial parameters such as enzyme turnover number (kcat), Michaelis constant (Km), and catalytic efficiency (kcat/Km or Vmax).

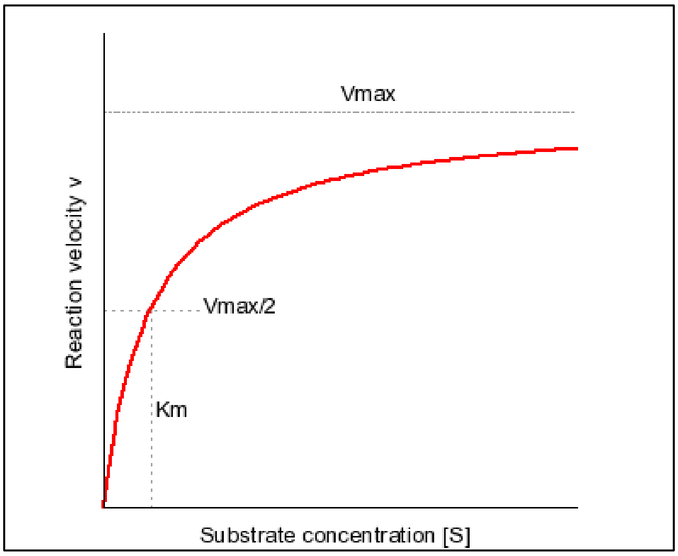

Figure 14: The reaction of an enzyme and substrate [25]

Km

The Km parameter reflects the affinity of an enzyme for its substrate. Lower Km values indicate higher affinity, meaning the enzyme can effectively bind to the substrate even at low concentrations. High Km values, on the other hand, suggest lower substrate affinity, requiring higher substrate concentrations to achieve half-maximal velocity.

Essentially, Km quantifies the enzyme-substrate binding strength, providing insights into the enzyme's efficiency in converting substrate into product. [25]

kcat

The kcat parameter, also known as the turnover number, represents the maximum number of substrate molecules converted to product per unit time by a single enzyme molecule when the enzyme is saturated with substrate. This can be represented using this formula: $kcat = \frac{V_{max}}{[Enzyme]}$, with [Enzyme] being the concentration of the enzyme in the enzyme-substrate reaction system.

It quantifies the catalytic efficiency of an enzyme, indicating how rapidly the enzyme can catalyse the conversion of substrate into product.

Enzymes with higher kcat values exhibit greater catalytic efficiency, as they can process substrates more rapidly. kcat is a fundamental parameter for comparing the catalytic capabilities of different enzymes and assessing their overall efficiency in catalysing biochemical reactions. [25]

Vmax

The Vmax parameter represents the maximum rate of a biochemical reaction catalysed by an enzyme when it is fully saturated with substrate, i.e., when all available enzyme active sites are bound to substrate molecules.

It reflects the intrinsic catalytic activity of the enzyme under optimal conditions, independent of substrate concentration. Vmax is achieved when the rate of the forward reaction (conversion of substrate to product) equals the rate of the reverse reaction (conversion of product back to substrate).

This parameter provides insights into the enzyme's overall catalytic capacity and serves as a reference point for comparing the catalytic efficiencies of different enzymes. [25]

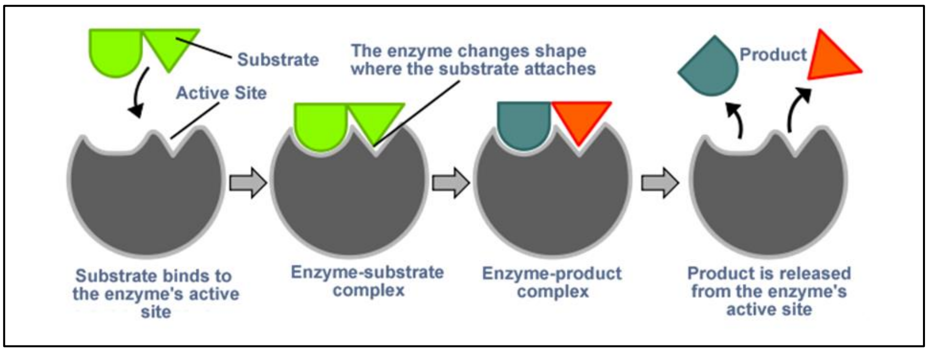

Figure 15: Enzyme and Substrate reaction [26]

The UniKP framework will be placed on Hugging Face as a model. An endpoint will be created from this endpoint to retrieve the Km, kcat and Vmax values. The UniKP component will send requests (enzyme and substrate) to this endpoint and receive the prediction of the kinetic parameters (Km, kcat, and Vmax).

The significance of predicting kcat, Km, and Vmax lies in the profound insights it offers into the functionality and substrate interactions of enzymes. These parameters are fundamental in understanding the efficiency and specificity of enzyme-substrate interactions, shedding light on the mechanisms underlying enzymatic processes.

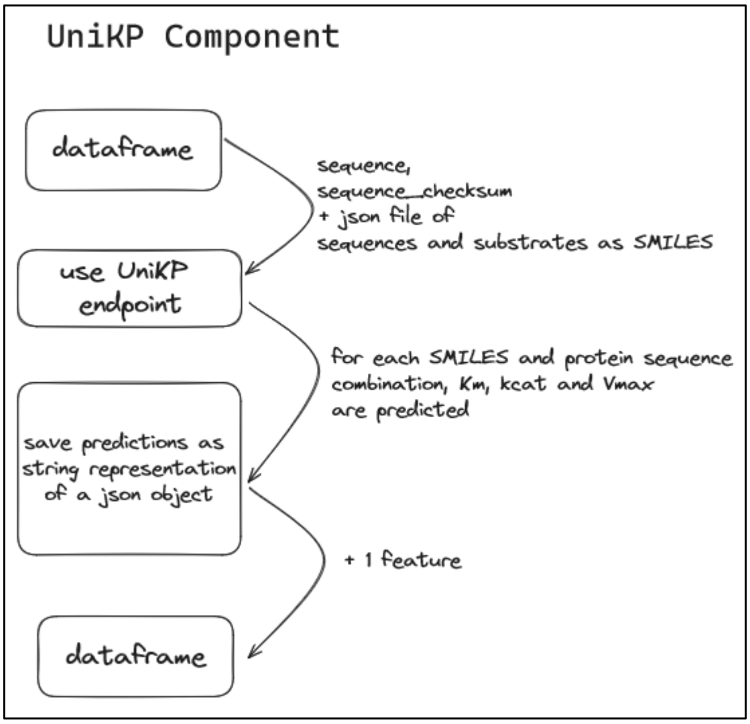

Figure 16: UniKP Component Structure

3.3 Processing the Fondant pipeline

The components can be linked together sequentially using Fondant. As mentioned in the Fondant section, this framework is used as a data pre-processing tool where each component does its own part in modifying the dataset provided. This section will visualize how these components are linked together in Fondant.

Components can be dependent on each other. A component that isn't listed here is the **generate checksum component**, which creates a new feature named sequence checksum. A checksum is a value calculated from a data set, typically a sequence of bytes or bits, for the purpose of verifying data integrity. It is commonly used in digital communication and data storage to detect errors that may have occurred during transmission or storage. The checksum is used in the pipeline as an id for the sequence.

The **generate checksum component** is placed in the beginning of the pipeline and the generated feature is used throughout the project. This means that if this component is removed it, depending on which component uses the checksum, the rest of the pipeline might not fully work. This also means that the **generate checksum component** can be swapped out with another checksum component that applies a different checksum function. The rest of the pipeline would still work in this case.

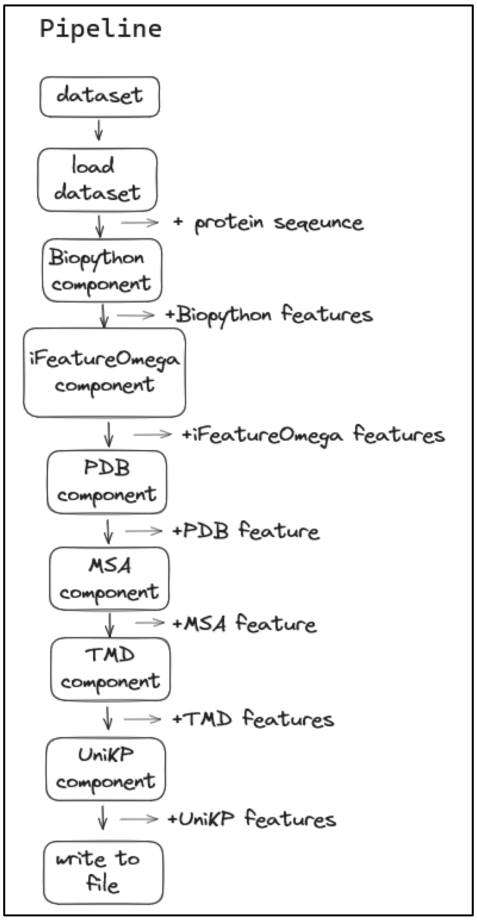

Figure 17: Pipeline Structure

3.4 Proof of Concept

This POC tries to establish correlations between kinetic parameters (Km, kcat, and Vmax) through the application of machine learning models and methodologies. Predicting the kcat parameter is particularly significant as it provides insights into enzyme catalytic efficiency. The interesting thing about kcat is whether it is possible to predict kcat using the components and features that are were calculated.

The POC aims to predict the kcat feature using various components as training features. This involves two distinct runs: one incorporating Km and Vmax features in the training data, and another without. This comparison serves to find out if the other kinetic parameters influence the accuracy of kcat prediction.

The entire code repository for this thesis is publicly available and can be found here: https://github.com/Detopall/bachelor-thesis-protein-feature-extraction

3.4.1 Data management and pre-processing

As mentioned before, the data is obtained from the UniProt site. Based upon the recommendation of my mentor, a specific type of protein family is used, namely CYP enzymes. The dataset consists of 2650 rows and 119 columns. The rows contain different protein sequences, and the columns contain the features provided by the feature extraction pipeline.

Not all components, mentioned in the previous sections, are used in the Proof of Concept. Only components that can be used in a regression model, so components that generate numerical features. Components such as the MSA component and the PDB component are not used. The TMD component also is not used for the simple fact that creating these features for the +2500 sequences would take an extremely long time.

These are the following components used in the pipeline:

- Biopython component
- iFeatureOmega component
- peptide component
- UniKP component

The POC contains a “clean-up” notebook, which performs multiple tasks to make sure that the data is ready for the regression models. Regression models cannot handle empty rows and it’s also not advisable to keep duplicates.

3.4.2 Introduction to regression models

Regression models are a type of statistical analysis used to estimate the relationship between a dependent variable and one or more independent variables. Regression models can be used to answer questions such as which variables have an impact on the dependent variable, and to what extent. They can also be used to make predictions based on the relationships between variables.

3.4.3 Polynomial Regression

Polynomial regression models the relationship between dependent and independent variables by fitting a polynomial function to the data. It is a nonlinear technique suitable for capturing complex relationships. The degree of the polynomial determines model complexity, with coefficients estimated via least squares regression. Challenges include overfitting, where the model becomes too complex, and underfitting, where it is too simplistic. To mitigate, careful degree selection and regularization techniques are crucial.

3.4.4 Random Forest

Random forest combines multiple decision trees to enhance model accuracy and stability. It's an ensemble method utilizing random subsets of data for each tree, then aggregating predictions. Suitable for classification and regression, it excels with high-dimensional data.

Challenges include overfitting and potential inefficiency. Addressing these requires careful selection of parameters like the number of trees and data subset size, alongside regularization techniques.

3.4.5 KNN

K-Nearest Neighbours classifies new data points based on similarity to existing ones. A simple and intuitive method, it's sensitive to the choice of neighbours and distance metric. Challenges include inefficiency and overfitting. Addressing these necessitates careful parameter selection and regularization techniques.

3.4.6 Gradient Boosting

Gradient boosting improves model accuracy by combining weak models in a sequence, each correcting the errors of its predecessor. It's effective for both classification and regression tasks, handling complex data well. Overfitting and inefficiency are common challenges, mitigated by careful selection of parameters like the number of trees and learning rate, along with regularization.

3.4.7 XGB (Extreme Gradient Boosting)

Extreme gradient boosting (XGB) is a highly efficient implementation of gradient boosting. Like gradient boosting, it builds a sequence of trees to correct errors iteratively. It's proficient with large datasets and employs regularization to combat overfitting. Overfitting and inefficiency remain challenges, addressed through parameter tuning and regularization techniques.

3.4.8 Train Test Split

The train-test-split principle in machine learning involves dividing the dataset into three subsets: a training set, a validation set, and a test set. The training set is used to train the model, the validation set is used to validate the model, and the test set is used to evaluate the model's performance on unseen data.

It's crucial to keep the validation set and test set separate to prevent knowledge leakage from the test set during hyperparameter tuning. This issue is addressed by using a training, test, and validation set. The validation set is utilized to fine-tune the model and compare different model types. Once a model is selected, the test set is used to assess the model's performance on new, unseen data.

This technique helps to ensure that the model can generalize well to new data and avoid overfitting to the training data.

3.4.9 Results

These are the results of each model with the training score and validation score. The results are split into two sections, with or without other kinetic parameters (Km and Vmax) in the training data. As mentioned before, the goal is to understand what influence the other kinetic parameters (Km and Vmax) have on the prediction of the kcat feature.

Table 1: Models used and their train and validation scores using the Km and Vmax in the training data

Model	**Train**	**Validation**
PolynomialRegressor	63.5%	0.0%
KNearestNeighbor	88.8%	76.0%
RandomForestRegressor	93.6%	80.9%
GradientBoostingRegressor	96.5%	81.9%
XGradientBoostingRegressor	100%	74.9%

The validation score is used to evaluate how good a model performs with unseen data. The highest validation score is met by using the GradientBoostingRegressor with a score of 80.9%. Gradient Boosting Regressor has been used in the POC to predict values on the test set.

The GBR using its best hyperparameters has managed to generate a score of 78% on the test dataset.

Table 2: Evaluation metrics scores using the Km and Vmax in the training data

Mean Absolute Error (MAE)	**Mean Squared Error (MSE)**	**Root Mean Squared Error (RMSE)**	**R^2**
0.478	0.634	0.796	0.780

The **MAE**, **MSE**, and **RMSE** indicate that, on average, the model's predictions are off by around 0.478 units. The **RMSE** of 0.796 suggests that the model's predictions have some spread around the actual values. The R-squared score of 0.780 suggests that the model explains a substantial portion of the variance in the target variable, indicating a reasonably good fit.

The following plot shows the predicted values of the model against the actual values, providing a visual representation of the model's accuracy. The goal is to minimize the difference between the predicted and actual values, indicating a better fit of the model. The closer the blue dots are to the middle line, the better the prediction.

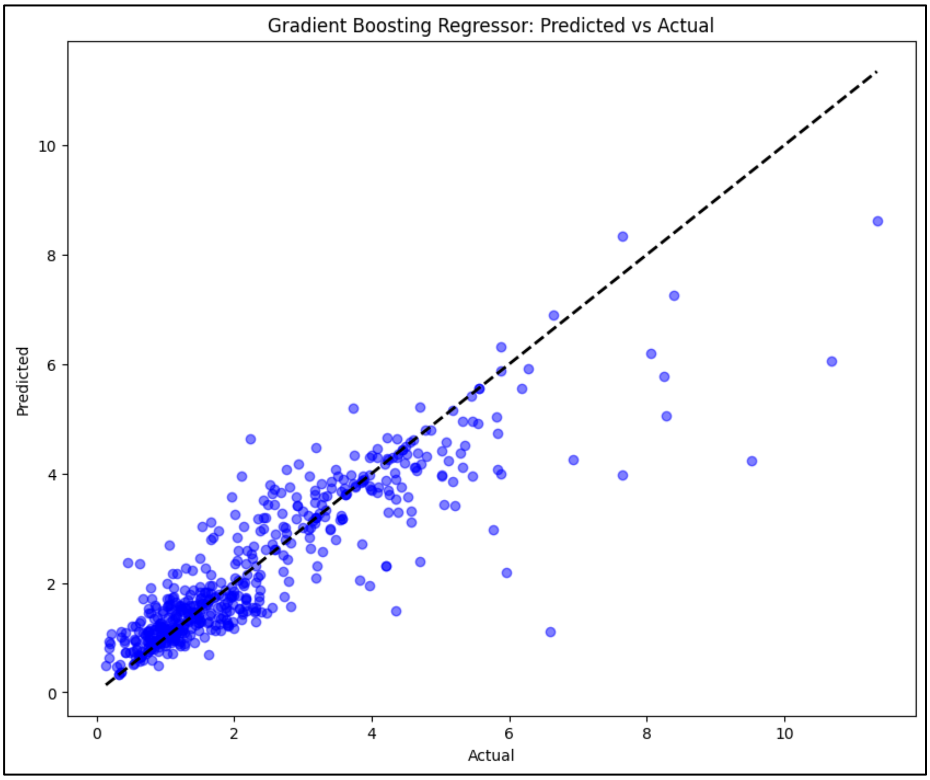

Figure 18: Predicted kcat values using GBR vs. the values taken from the UniKP model with other kinetic parameters present

There are more dots in the beginning of the graph, indicating that most of the kcat values are between 0 and 4. This has to do with the protein sequences and substrate used to predict the kcat values. Other protein sequences or substrates will show different results.

Table 3: Evaluation metrics scores without using the Km and Vmax in the training data

Model	**Train**	**Validation**
PolynomialRegressor	63.0%	0.0%
KNearestNeighbor	89.0%	75.5%
RandomForestRegressor	93.6%	81.0%
GradientBoostingRegressor	96.8%	81.1%
XGradientBoostingRegressor	100%	75.1%

The highest validation score is met by using the Gradient Boosting Regressor with a score of 81.1%. Gradient Boosting Regressor has been used also in this section of the POC to predict values on the test set. The Random Forest Regressor will not be chosen, because it lacks the capacity to be more complex in future implementations.

The GBR using its best hyperparameters has managed to generate a score of 78% on the test dataset. This is the exact same as the model containing the other kinetic parameters, Km and Vmax.

Table 4: Evaluation metrics scores without using the Km and Vmax in the training data

Mean Absolute Error (MAE)	**Mean Squared Error (MSE)**	**Root Mean Squared Error (RMSE)**	R^2
0.483	0.630	0.796	0.781

The **MAE**, **MSE**, and **RMSE** indicate that, on average, the model's predictions are off by around 0.483 units. The **RMSE** of 0.796 suggests that the model's predictions have some spread around the actual values. The R-squared score of 0.781 suggests that the model explains a substantial portion of the variance in the target variable, indicating a reasonably good fit.

The following plot shows the predicted values of the model against the actual values, providing a visual representation of the model's accuracy. The goal is to minimize the difference between the predicted and actual values, indicating a better fit of the model. The closer the blue dots are to the middle line, the better the prediction.

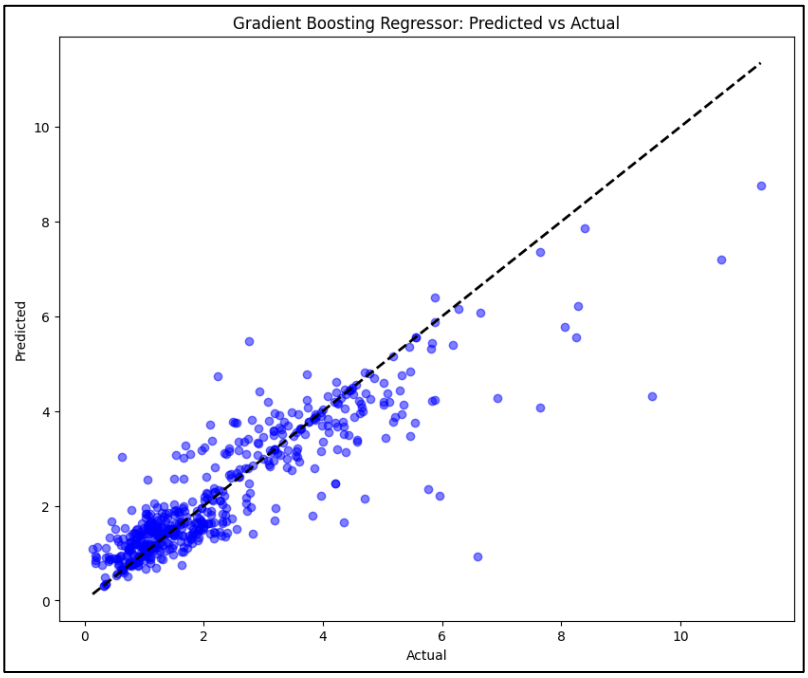

Figure 19: Predicted kcat values using GBR vs. the values taken from the UniKP model without other kinetic parameters present

There are more dots in the beginning of the graph, indicating that most of the kcat values are between 0 and 4. This has to do with the protein sequences and substrate used to predict the kcat values. Other protein sequences or substrates will show different results.

3.5 Feature importance

Feature importance in machine learning refers to the process of determining the relative importance of each feature in a dataset when building a model. This is useful for understanding which features contribute the most to the final prediction, and can be used for feature selection, model interpretability, model debugging, business decision-making, and improving model performance.

The target feature, kcat the kinetic enzyme parameter, will be used to calculate the feature importance. This has been done using an Extreme Gradient Boosting Regression model trained on training data of the dataset. The numerical columns except the target feature (X) will be fitted with the target feature (Y). The goal is to find the best possible fit between the model and the data, so that the model can accurately predict future outcomes based on the patterns and relationships found in the data.

The figure below shows the feature importance generated by the Extreme Gradient Boosting Regression model. The more important a feature, the more influence this feature has on the target feature.

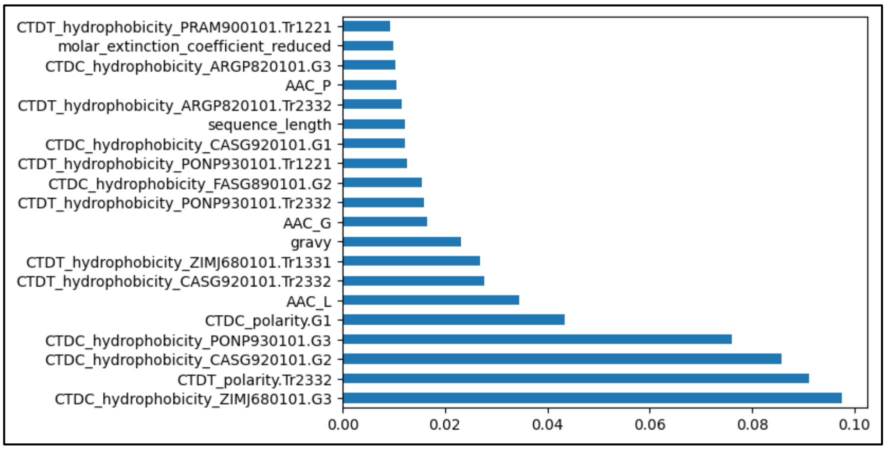

Figure 20: Feature importance of kcat with other kinetic parameters present

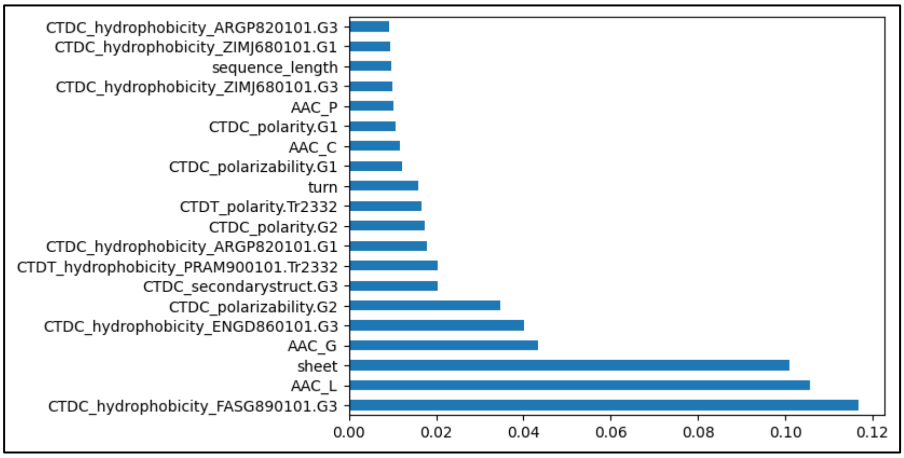

Figure 21: Feature importance of kcat without other kinetic parameters present

While Figure 14 does contain the Km and Vmax features in the dataset, the model doesn't find the Km or Vmax feature important enough, while being both kinetic parameters such as the target feature, kcat. The Km and Vmax features have a feature importance of 0.0039 and 0.0036 respectively.

There is also a very noticeable influence of the CTDC descriptor. The CTDC descriptor focuses on hydrophobicity, categorizing amino acids into polar, neutral, and hydrophobic groups to calculate the global composition of these residues in a protein. The naming of these features is created using iFeatureOmega. Each amino acid is part of a certain group. The most influential feature "CTDC_ hydrophobicity_FASG890101.G3" can be decoded as follows:

- CTDC: refers to the descriptor chosen (Composition).
- hydrophobicity_FASG890101: indicates that this feature is related to the hydrophobicity property using the FASG890101 scale.
- .G3: indicates that this feature is derived from Group 3 of amino acids. Group 3 referring to the different groups within the code of iFeatureOmega.

[27]

3.6 Univariate Linear Regression

The univariate linear regression per feature calculates the beta values for each feature, which represent the contribution of each independent variable (every feature other than kcat) toward explaining the dependent variable (kcat). The most influential features are the top 20 features with the highest absolute beta values, indicating a stronger linear relationship with the dependent variable. The following figures will show the results from these calculations.

Table 5: Top 20 most influential features using univariate linear regression including Km and Vmax

Feature	Beta
AAC_G	0.673600
CTDC_hydrophobicity_FASG890101.G2	0.640221
CTDT_polarity.Tr2332	0.627819
AAC_L	-0.627045
CTDT_hydrophobicity_FASG890101.Tr1221	0.622134
CTDC_polarizability.G1	0.607265
CTDT_hydrophobicity_PONP930101.Tr1221	0.606291
CTDC_hydrophobicity_ARGP820101.G1	0.603623
CTDC_hydrophobicity_ZIMJ680101.G3	-0.597512
CTDC_hydrophobicity_ARGP820101.G3	-0.595664
km	0.594058
CTDT_hydrophobicity_ZIMJ680101.Tr1331	-0.581981
CTDC_secondarystruct.G3	0.575814
CTDT_hydrophobicity_PRAM900101.Tr1221	0.573320
turn	0.569751
CTDC_hydrophobicity_PONP930101.G3	-0.554816
CTDC_polarity.G1	-0.554816
CTDT_hydrophobicity_ENGD860101.Tr1221	0.552128
CTDC_polarizability.G2	-0.544741
CTDC_hydrophobicity_ZIMJ680101.G1	0.530746

This table shows in the **km** feature in the middle of the table with a beta value of 0.59. This is significantly lower than the **AAC_G** feature, which would normally not have such a high influence on the kcat feature at first glance. The feature **vmax** isn't even in the top 20, it has a beta value of 0.52.

Table 6: Top 20 most influential features using univariate linear regression without other kinetic parameters

Feature	Beta
AAC_G	0.674959
CTDC_hydrophobicity_FASG890101.G2	0.639354
CTDT_polarity.Tr2332	0.626229
AAC_L	-0.624828
CTDT_hydrophobicity_FASG890101.Tr1221	0.620793
CTDC_polarizability.G1	0.605700
CTDT_hydrophobicity_PONP930101.Tr1221	0.604607
CTDC_hydrophobicity_ARGP820101.G1	0.601889
CTDC_hydrophobicity_ZIMJ680101.G3	-0.595518

CTDC_hydrophobicity_ARGP820101.G3	-0.593629
CTDT_hydrophobicity_ZIMJ680101.Tr1331	-0.580780
CTDC_secondarystruct.G3	0.574316
CTDT_hydrophobicity_PRAM900101.Tr1221	0.572036
turn	0.568280
CTDC_polarity.G1	-0.551907
CTDC_hydrophobicity_PONP930101.G3	-0.551907
CTDT_hydrophobicity_ENGD860101.Tr1221	0.551302
CTDC_polarizability.G2	-0.543307
CTDC_hydrophobicity_ZIMJ680101.G1	0.527586
CTDC_hydrophobicity_ENGD860101.G2	0.525385

This table doesn't contain any kinetic parameters, but again the amino acid compositions of amino acids G and L are at the topmost influential features for the kcat feature.

3.7 Discussion and Implications

The POC has provided valuable information into the application of regression models for predicting kinetic enzyme parameters based on protein sequence data. Throughout these finding some implications have emerged.

The **km** feature is visible in the top 20 beta values, but the **vmax** feature is not. They do not really pose a significant positive influence on the model, with both being around 0.59 and 0.52 respectively. Positive meaning that the higher the value of **vmax** and **km**, the higher the value of the target variable. What is interesting is that there are pretty "random" variables present in the top 20 beta values. Features such as **AAC_G** and **AAC_L**, these are the amino acid compositions of the **G** and **L** are amino acids. This means that the amount of **G** and **L** amino acids in the sequence has a significantly greater influence on the target variable than the kinetic parameters **vmax** and **km**. Taking the previous statements into account, it is safe to assume that there needs to be more research around the influence of feature on kinetic parameters.

The regression models exhibited varying performance levels. The Gradient Boosting Regressor was chosen as the final model, but the Random Forest Regression model also had a high validation accuracy. Considering future scalability and complexity, Gradient Boosting was chosen for its ability to handle intricate relationships and maintain performance integrity. This decision aligns with the goal of developing a robust and adaptable model for real-world scenarios.

4 Conclusion

Protein engineering is a critical tool for developing sustainable production methods to address the pressing challenges of climate change and environmental degradation. By optimizing enzymes to enhance the performance of microbial cell factories, protein engineering enables the conversion of renewable feedstocks into high-value products in an eco-friendlier manner compared to traditional fossil fuel-based production. Calculating key enzyme features is essential for guiding the design of high-performance enzymes tailored for specific biotechnology applications.

Feature extraction from protein sequences can be accomplished through various approaches. This thesis investigated the utilization of a pipeline comprising multiple components to extract features, leveraging open-source AI models and functions. The chosen tool was Fondant, a project developed by ML6. Fondant was specifically selected for its modular design and flexibility in applying different components.

With this tool and other open-source packages, models, programs and other resources, the following components were created:

- Biopython Component
- iFeatureOmega Component
- PDB Components
- Peptide Component
- MSA Component
- TMD Component
- UniKP Component

One of the significant findings of this research was the successful development of a model capable of predicting the kcat parameters of a protein sequence and substrate combination using the components created within the Fondant pipeline. The effectiveness of the model, which utilized the Gradient Boosting Regressor trained on numerous protein sequences, further highlights the potential of this approach in accurately predicting kcat parameters.

Based on these conclusions, it is recommended that future researchers and practitioners consider adding more components or refactoring existing components with the newest models if applicable. Building upon the existing pipeline and continuing to optimize feature extraction techniques will be crucial for the success of the project.

This research and thesis also implemented the UniKP framework, a crucial component for feature extraction of kinetic parameters. By incorporating the UniKP framework, this research has demonstrated the ability to adapt and integrate advanced bioinformatics tools to address complex research questions. This implementation not only bridges the gap between theoretical knowledge and practical application but also paves the way for further research and development in protein sequence analysis.

It also must be noted that when observing the results of the POC, the kinetic parameters of Km and Vmax did not seem to have a grave influence on the prediction of the kcat parameter. On the other hand, the CDTC and amino acids G and L had an immense on the target feature. This must be researched further to understand the underlying influence of features on kinetic parameters.

Conclusion

In response to the urgent need for sustainable production methods amidst climate change, industrial biotechnology has emerged as a promising solution. This thesis introduces a pipeline for automating enzyme feature calculations to expedite the discovery of high-performance enzymes for sustainable production. By leveraging advanced tools for protein feature analysis, this pipeline aims to optimize enzyme design for valuable compound production like flavonoids and terpenoids. Future research could focus on broadening the pipeline's applicability to diverse biotechnology areas and exploring machine learning for enhanced enzyme design. This work contributes to advancing sustainable production methods and underscores the potential for innovative enzyme optimization in industrial biotechnology.

AI Engineering Prompts

AI models have been used to create this thesis. This section will contain a link of the conversation with ChatGPT and Perplexity.

https://chat.openai.com/share/4e4db43f-0aac-4d7e-9509-7d5768a88b52

https://www.perplexity.ai/search/Problem-and-scope-vUGqLTn6Qm29GCph0QviBg

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